WoSM Academy

MAT231-F27 · Calculus II

Second-order linear equations

Constant coefficients, oscillations and resonance

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Lecture 24 · Unit 24 · Fall 2027

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Last time

- Initial value problems $y' = f(t, y)$; existence and uniqueness; direction fields; Euler's method.
- Separable equations: $\int dy/h(y) = \int g(t)\,dt$.
- Linear equations $y' + py = q$: the integrating factor $e^{\int p}$; solutions are particular plus homogeneous.

Newton's law $F = ma$ involves the second derivative of position. Today: second-order linear equations with constant coefficients — springs, circuits, and anything that oscillates.

This lecture

01 Second-order linear equations

02 The characteristic equation

03 Oscillations

04 Forced oscillations

05 Summary and practice

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 7.1–7.3.

01

Second-order linear equations

A mass on a spring

A mass m on a spring with stiffness k, a damper with coefficient $c \geq 0$, and an external force $F(t)$. With $x(t)$ the displacement from equilibrium, Newton's law gives

$$m\,x'' = \underbrace{-kx}_{\text{spring}}\ \underbrace{-\,c\,x'}_{\text{damping}} + F(t), \qquad \text{that is} \qquad mx'' + cx' + kx = F(t).$$

- An electric circuit with inductance L, resistance R and capacitance C gives the same equation for the charge: $Lq'' + Rq' + \frac{1}{C}q = E(t)$.
- In general: $ay'' + by' + cy = g(t)$, with constants $a \neq 0$, b, c. It is **homogeneous** if $g = 0$.

The space of solutions

Theorem

For any t_0, y_0, v_0, the equation $ay'' + by' + cy = g(t)$ (with g continuous) has exactly one solution with $y(t_0) = y_0$ and $y'(t_0) = v_0$, defined for all t.

Two initial conditions for a second-order equation: position and velocity.

- **Superposition**: if y_1, y_2 solve the homogeneous equation, so does $C_1y_1 + C_2y_2$.
- The homogeneous solutions form a vector space of **dimension** 2: the map $y \mapsto (y(0), y'(0))$ is linear, and by the theorem a bijection onto $\mathbb{R}^2$.
- So if y_1, y_2 are two independent solutions, every solution is $C_1y_1 + C_2y_2$. They are independent if the **Wronskian** $W = y_1y_2' - y_1'y_2$ is not 0.

02

The characteristic equation

Try an exponential

For $ay'' + by' + cy = 0$, try $y = e^{rt}$:

$$ar^2e^{rt} + bre^{rt} + ce^{rt} = (ar^2 + br + c)\,e^{rt} = 0.$$

So e^{rt} is a solution exactly when r solves the **characteristic equation**

$$ar^2 + br + c = 0.$$

Three cases, according to the discriminant $b^2 - 4ac$:

- > 0: two real roots $r_1 \neq r_2$;
- $= 0$: one double root r;
- < 0: two complex roots $\alpha \pm i\beta$.

In each case: find the roots, read off two independent solutions, and fit C_1, C_2 to the initial conditions.

Two real roots

Two real roots $r_1 \neq r_2$

$$y = C_1 e^{r_1 t} + C_2 e^{r_2 t}.$$

The two solutions are independent: their Wronskian is $(r_2 - r_1)\, e^{(r_1+r_2)t} \neq 0$.

Example $(y'' - y' - 2y = 0,\ \ y(0) = 1,\ y'(0) = 2)$

$r^2 - r - 2 = (r-2)(r+1)$, so $y = C_1 e^{2t} + C_2 e^{-t}$.
$C_1 + C_2 = 1$ and $2C_1 - C_2 = 2$ give $C_1 = 1$, $C_2 = 0$: $\ y = e^{2t}$.

A double root

A double root $r = -\frac{b}{2a}$

$$y = (C_1 + C_2 t)\, e^{rt}.$$

Why te^{rt} is a solution: with $p(r) = ar^2 + br + c$, a short computation gives

$$a\,(te^{rt})'' + b\,(te^{rt})' + c\,te^{rt} = p(r)\,te^{rt} + p'(r)\,e^{rt},$$

and a double root has $p(r) = p'(r) = 0$.

Example $(y'' + 2y' + y = 0)$

$r^2 + 2r + 1 = (r+1)^2$: $\;y = (C_1 + C_2 t)\, e^{-t}$.

Complex exponentials

The Taylor series of e^x (Lecture 6) also converges for complex x. With $x = i\theta$, and $i^2 = -1$, the even and odd terms separate:

$$e^{i\theta} = \Big(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\Big) + i\Big(\theta - \frac{\theta^3}{3!} + \dots\Big) = \cos\theta + i\sin\theta.$$

- This is **Euler's formula**. So $e^{(\alpha+i\beta)t} = e^{\alpha t}(\cos\beta t + i\sin\beta t)$, and its derivative is $(\alpha + i\beta)\,e^{(\alpha+i\beta)t}$, by the product rule.
- If $r = \alpha + i\beta$ is a root of $ar^2 + br + c$, then e^{rt} is a complex solution. Since a, b, c are real, its real and imaginary parts are real solutions:

$$e^{\alpha t}\cos\beta t \qquad \text{and} \qquad e^{\alpha t}\sin\beta t.$$

Complex roots

Two complex roots $\alpha \pm i\beta$ ($\beta \neq 0$)

$y = e^{\alpha t}\big(C_1 \cos \beta t + C_2 \sin \beta t\big)$.

Example ($y'' + 4y = 0$, $y(0) = 1$, $y'(0) = 2$)

$r = \pm 2i$: $y = C_1 \cos 2t + C_2 \sin 2t$. $y(0) = C_1 = 1$, $y'(0) = 2C_2 = 2$: $y = \cos 2t + \sin 2t$.

Example ($y'' + 2y' + 5y = 0$: an oscillation that dies out)

$r = -1 \pm 2i$: $y = e^{-t}\big(C_1 \cos 2t + C_2 \sin 2t\big)$.

03

Oscillations

Free undamped oscillation

$mx'' + kx = 0$, that is $x'' + \omega_0^2 x = 0$ with $\omega_0 = \sqrt{k/m}$. Roots $\pm i\omega_0$:

$$x(t) = C_1 \cos \omega_0 t + C_2 \sin \omega_0 t = A \cos(\omega_0 t - \delta),$$

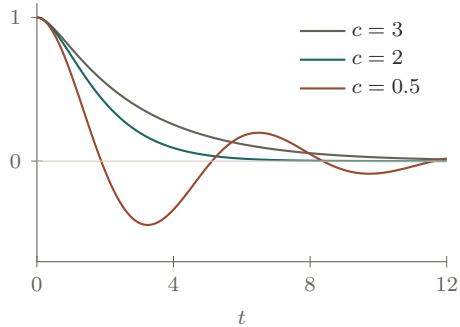
with **amplitude** $A = \sqrt{C_1^2 + C_2^2}$ and phase δ ($\cos \delta = C_1/A$, $\sin \delta = C_2/A$).

- Simple harmonic motion with **natural frequency** ω_0 and period $2\pi/\omega_0$.
- A stiffer spring (larger k) or lighter mass (smaller m) oscillates faster.
- The energy $\frac{1}{2}m(x')^2 + \frac{1}{2}kx^2$ is constant: differentiate it and use the equation.

Damping

$mx'' + cx' + kx = 0$ with $c > 0$: roots $\dfrac{-c \pm \sqrt{c^2 - 4mk}}{2m}$, both with negative real part.

- $c^2 > 4mk$: **overdamped**. Two negative real roots: a slow return, no oscillation.
- $c^2 = 4mk$: **critically damped**: the fastest return without overshoot.
- $c^2 < 4mk$: **underdamped**. A decaying oscillation $e^{-ct/2m}\cos(\mu t - \delta)$.



$m = k = 1,\ \ x(0) = 1,\ x'(0) = 0.$

04

Forced oscillations

Nonhomogeneous equations

Theorem

The general solution of $ay'' + by' + cy = g(t)$ is $y = y_p + C_1 y_1 + C_2 y_2$: one particular solution plus the general homogeneous solution. (The difference of two solutions solves the homogeneous equation.)

Undetermined coefficients: guess y_p of the same form as g.

$g(t)$	try y_p
polynomial of degree n	polynomial of degree n
e^{kt}	Ae^{kt}
$\cos \omega t$ or $\sin \omega t$	$A \cos \omega t + B \sin \omega t$

If the guess solves the homogeneous equation, multiply it by t (again, if necessary).

Examples

Example $(y'' - y' - 2y = 4t)$

Try $y_p = At + B$: $0 - A - 2(At + B) = 4t$, so $A = -2$ and $B = 1$. With the homogeneous solutions from before:

$$y = C_1 e^{2t} + C_2 e^{-t} - 2t + 1.$$

Example $(y'' + 4y = 3\cos t)$

Try $y_p = A\cos t$ (no $\sin t$ is needed, as there is no y' term): $-A + 4A = 3$, so $A = 1$:

$$y = \cos t + C_1 \cos 2t + C_2 \sin 2t.$$

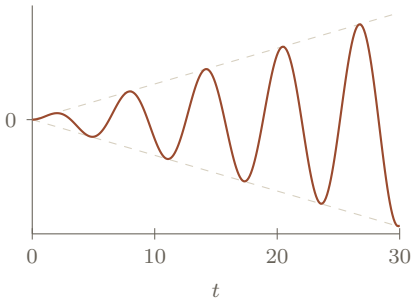
Resonance

$x'' + \omega_0^2 x = \cos \omega t$: a spring pushed at frequency ω.

- $\omega \neq \omega_0$: $x_p = \dfrac{\cos \omega t}{\omega_0^2 - \omega^2}$, which grows as $\omega \to \omega_0$.
- $\omega = \omega_0$: the guess $A \cos \omega_0 t$ solves the homogeneous equation; multiplying by t gives

$$x_p = \frac{t \sin \omega_0 t}{2\omega_0},$$

which grows without bound: **resonance**.



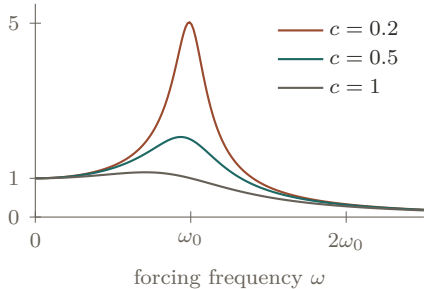
$x'' + x = \cos t, \;\; x_p = \frac{1}{2} t \sin t.$

Damping keeps the response bounded

$mx'' + cx' + kx = \cos \omega t$ with $c > 0$ has a solution $A \cos \omega t + B \sin \omega t$ of amplitude

$$\frac{1}{\sqrt{(k - m\omega^2)^2 + c^2\omega^2}}.$$

- The homogeneous part decays: every solution approaches this **steady state**.
- It peaks near $\omega_0 = \sqrt{k/m}$, higher for smaller c.



Amplitude, $m = k = 1$.

05

Summary and practice

Summary

- $ay'' + by' + cy = 0$: solutions form a 2-dimensional space; two initial conditions fix one.
- Characteristic equation $ar^2 + br + c = 0$. Real roots: $C_1 e^{r_1 t} + C_2 e^{r_2 t}$; double root: $(C_1 + C_2 t)e^{rt}$; complex roots $\alpha \pm i\beta$: $e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t)$, by Euler's formula.
- Springs: simple harmonic motion; over-, critically and underdamped motion.
- Forced: $y = y_p + y_h$; undetermined coefficients; resonance when the forcing matches the natural frequency.

Practice

1. Solve $y'' - 5y' + 6y = 0$, $y(0) = 1$, $y'(0) = 0$.
2. Solve $y'' - 4y' + 4y = 0$, $y(0) = 1$, $y'(0) = 0$.
3. Find the general solution of $y'' + 2y' + 10y = 0$.
4. A mass $m = 1$ hangs on a spring with $k = 4$. For which damping coefficients c is the motion underdamped, critically damped, overdamped?
5. Find the general solution of $y'' + y' - 2y = 2e^{3t}$.
6. Find a particular solution of $y'' + 9y = \sin 3t$.
7. Show that the Wronskian of $e^{r_1 t}$ and $e^{r_2 t}$ is $(r_2 - r_1)\, e^{(r_1 + r_2)t}$.

Answers

1. $r = 2, 3$: $C_1 + C_2 = 1$, $2C_1 + 3C_2 = 0$: $y = 3e^{2t} - 2e^{3t}$.
2. $r = 2$ (double): $y = (1 + C_2 t)e^{2t}$, $y'(0) = 2 + C_2 = 0$: $y = (1 - 2t)\, e^{2t}$.
3. $r = -1 \pm 3i$: $y = e^{-t}(C_1 \cos 3t + C_2 \sin 3t)$.
4. $c^2 - 16$: underdamped for $c < 4$, critical for $c = 4$, overdamped for $c > 4$.
5. $r = 1, -2$; $y_p = Ae^{3t}$ with $(9 + 3 - 2)A = 2$: $y = C_1 e^t + C_2 e^{-2t} + \frac{1}{5} e^{3t}$.
6. Resonance: try $t(A \cos 3t + B \sin 3t)$; $y_p = -\frac{t \cos 3t}{6}$.
7. $W = e^{r_1 t} \cdot r_2 e^{r_2 t} - r_1 e^{r_1 t} \cdot e^{r_2 t} = (r_2 - r_1)\, e^{(r_1 + r_2)t}$.

Next time

Lecture 25 — Systems of linear differential equations (Unit 25)

- $\mathbf{x}' = A\mathbf{x}$: solutions from eigenvalues and eigenvectors
- Phase portraits: saddles, nodes, spirals; stability
- The matrix exponential, and a look back at the module

Reading: Apostol, *Calculus, Volume 2*, Chapter 7.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.